

Name – Ganesh Malaji Ghule
Roll No - 37

Assignment No 3

Jupyter Practical 3 (autosaved)  Logout

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In [1]: `import pandas as pd`

In [2]: `data = pd.read_csv("loan_data.csv")`
`data`

Out[2]:

	Loan_ID	Gender	Age	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_His
0	LP001002	Male	35.0	No	0	Graduate	No	5849	0.0	NaN	360.0	
1	LP001003	Male	NaN	Yes	1	Graduate	No	4583	1508.0	128.0	360.0	
2	LP001005	Male	25.0	Yes	0	Graduate	Yes	3000	0.0	66.0	360.0	
3	LP001006	Male	26.0	Yes	0	Not Graduate	No	2583	2358.0	120.0	360.0	
4	LP001008	Male	30.0	No	0	Graduate	No	6000	0.0	141.0	360.0	
...	...	...	...	...	...	...	...	...	...	...	...	...
609	LP002978	Female	25.0	No	0	Graduate	No	2900	0.0	71.0	360.0	
610	LP002979	Male	50.0	Yes	3+	Graduate	No	4106	0.0	40.0	180.0	
611	LP002983	Male	38.0	Yes	1	Graduate	No	8072	240.0	253.0	360.0	
612	LP002984	Male	45.0	Yes	2	Graduate	No	7583	0.0	187.0	360.0	
613	LP002990	Female	26.0	No	0	Graduate	Yes	4583	0.0	133.0	360.0	

614 rows × 14 columns

In [3]: `data.head()`

Out[3]:

	Loan_ID	Gender	Age	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_His
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In [3]: `data.head()`

Out[3]:

	Loan_ID	Gender	Age	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_His
0	LP001002	Male	35.0	No	0	Graduate	No	5849	0.0	NaN	360.0	
1	LP001003	Male	NaN	Yes	1	Graduate	No	4583	1508.0	128.0	360.0	
2	LP001005	Male	25.0	Yes	0	Graduate	Yes	3000	0.0	66.0	360.0	
3	LP001006	Male	26.0	Yes	0	Not Graduate	No	2583	2358.0	120.0	360.0	
4	LP001008	Male	30.0	No	0	Graduate	No	6000	0.0	141.0	360.0	

In [4]: `data.tail()`

Out[4]:

	Loan_ID	Gender	Age	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_His
609	LP002978	Female	25.0	No	0	Graduate	No	2900	0.0	71.0	360.0	
610	LP002979	Male	50.0	Yes	3+	Graduate	No	4106	0.0	40.0	180.0	
611	LP002983	Male	38.0	Yes	1	Graduate	No	8072	240.0	253.0	360.0	
612	LP002984	Male	45.0	Yes	2	Graduate	No	7583	0.0	187.0	360.0	
613	LP002990	Female	26.0	No	0	Graduate	Yes	4583	0.0	133.0	360.0	

In [5]: `data.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 614 entries, 0 to 613
Data columns (total 14 columns):
#   Column              Non-Null Count  Dtype
---  -

```

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Column Non-Null Count Dtype

0 Loan_ID 614 non-null object
1 Gender 601 non-null object
2 Age 513 non-null float64
3 Married 611 non-null object
4 Dependents 599 non-null object
5 Education 614 non-null object
6 Self_Employed 582 non-null object
7 ApplicantIncome 614 non-null int64
8 CoapplicantIncome 614 non-null float64
9 LoanAmount 592 non-null float64
10 Loan_Amount_Term 600 non-null float64
11 Credit_History 564 non-null float64
12 Property_Area 614 non-null object
13 Loan_Status 614 non-null object
dtypes: float64(5), int64(1), object(8)
memory usage: 67.3+ KB

In [6]: data.describe()

Out[6]:

	Age	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History
count	513.000000	614.000000	614.000000	592.000000	600.000000	564.000000
mean	32.101365	5403.459283	1621.245798	146.412162	342.000000	0.842199
std	7.732178	6109.041673	2926.248369	85.587325	65.12041	0.364878
min	24.000000	150.000000	0.000000	9.000000	12.000000	0.000000
25%	25.000000	2877.500000	0.000000	100.000000	360.000000	1.000000

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Column Non-Null Count Dtype

0 Loan_ID 614 non-null object
1 Gender 601 non-null object
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6 Self_Employed 582 non-null object
7 ApplicantIncome 614 non-null int64
8 CoapplicantIncome 614 non-null float64
9 LoanAmount 592 non-null float64
10 Loan_Amount_Term 600 non-null float64
11 Credit_History 564 non-null float64
12 Property_Area 614 non-null object
13 Loan_Status 614 non-null object
dtypes: float64(5), int64(1), object(8)
memory usage: 67.3+ KB

In [6]: data.describe()

Out[6]:

	Age	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History
count	513.000000	614.000000	614.000000	592.000000	600.000000	564.000000
mean	32.101365	5403.459283	1621.245798	146.412162	342.000000	0.842199
std	7.732178	6109.041673	2926.248369	85.587325	65.12041	0.364878
min	24.000000	150.000000	0.000000	9.000000	12.000000	0.000000
25%	25.000000	2877.500000	0.000000	100.000000	360.000000	1.000000
50%	30.000000	3812.500000	1188.500000	128.000000	360.000000	1.000000
75%	38.000000	5795.000000	2297.250000	168.000000	360.000000	1.000000
max	56.000000	81000.000000	41667.000000	700.000000	480.000000	1.000000

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25% 25.000000 2877.500000 0.000000 100.000000 360.000000 1.000000
50% 30.000000 3812.500000 1188.500000 128.000000 360.000000 1.000000
75% 38.000000 5795.000000 2297.250000 168.000000 360.000000 1.000000
max 56.000000 81000.000000 41667.000000 700.000000 480.000000 1.000000

```
In [7]: data.isnull().sum()
```

```
Out[7]: Loan_ID      0  
Gender      13  
Age       101  
Married      3  
Dependents   15  
Education     0  
Self_Employed 32  
ApplicantIncome 0  
CoapplicantIncome 0  
LoanAmount    22  
Loan_Amount_Term 14  
Credit_History 50  
Property_Area  0  
Loan_Status    0  
dtype: int64
```

```
In [8]: mean = data.mean(numeric_only=True)  
mean
```

```
Out[8]: Age      32.101365  
ApplicantIncome 5403.459283  
CoapplicantIncome 1621.245798  
LoanAmount     146.412162  
Loan_Amount_Term 342.000000  
Credit_History  0.842199  
dtype: float64
```

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```
In [9]: data['LoanAmount'].mean()
```

```
Out[9]: 146.41216216216216
```

```
In [10]: data['Loan_Amount_Term'].mean()
```

```
Out[10]: 342.0
```

```
In [11]: median = data.median(numeric_only=True)  
median
```

```
Out[11]: Age      30.0  
ApplicantIncome 3812.5  
CoapplicantIncome 1188.5  
LoanAmount     128.0  
Loan_Amount_Term 360.0  
Credit_History  1.0  
dtype: float64
```

```
In [12]: data['Age'].median()
```

```
Out[12]: 30.0
```

```
In [13]: minimum = data.min(numeric_only=True)  
minimum
```

```
Out[13]: Age      24.0  
ApplicantIncome 150.0  
CoapplicantIncome 0.0  
LoanAmount      9.0  
Loan_Amount_Term 12.0  
Credit_History  0.0  
dtype: float64
```

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```
CoapplicantIncome    0.0
LoanAmount           9.0
Loan_Amount_Term     12.0
Credit_History       0.0
dtype: float64
```

```
In [14]: maximum = data.max(numeric_only=True)
maximum
```

```
Out[14]: Age                56.0
ApplicantIncome         81000.0
CoapplicantIncome       41667.0
LoanAmount              700.0
Loan_Amount_Term        480.0
Credit_History          1.0
dtype: float64
```

```
In [15]: std = data.std(numeric_only=True)
std
```

```
Out[15]: Age                7.732178
ApplicantIncome         6109.041673
CoapplicantIncome       2926.248369
LoanAmount              85.587325
Loan_Amount_Term        65.120410
Credit_History          0.364878
dtype: float64
```

```
In [16]: data['Age'].std()
```

```
Out[16]: 7.732178229043358
```

```
In [17]: data.groupby('Age').count()
```

```
Out[17]:
```

Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_Histo
---------	--------	---------	------------	-----------	---------------	-----------------	-------------------	------------	------------------	--------------

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```
In [17]: data.groupby('Age').count()
```

```
Out[17]:
```

Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_Histo
24.0	22	21	21	19	22	20	22	22	21	21
25.0	107	105	107	105	107	102	107	107	102	104
26.0	87	86	87	87	87	81	87	87	86	86
27.0	23	23	22	22	23	22	23	23	22	21
28.0	4	4	4	4	4	2	4	4	3	3
30.0	53	53	53	52	53	52	53	53	51	52
31.0	23	22	23	20	23	23	23	23	22	23
32.0	18	18	18	18	18	18	18	18	17	17
35.0	7	7	7	6	7	7	7	7	6	6
37.0	18	17	18	18	18	16	18	18	17	18

```
In [18]: import numpy as np
```

```
In [19]: data=pd.read_csv("iris.csv")
data
```

```
Out[19]:
```

Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	Iris-setosa
1	2	4.9	3.0	1.4	Iris-setosa
2	3	4.7	3.2	1.3	Iris-setosa

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In [20]: `data.groupby('Species').count()`

Out[20]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Species					
Iris-setosa	50	50	50	50	50
Iris-versicolor	50	50	50	50	50
Iris-virginica	50	50	50	50	50

In [21]: `data.groupby('Species').mean()`

Out[21]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Species					
Iris-setosa	25.5	5.006	3.418	1.464	0.244
Iris-versicolor	75.5	5.936	2.770	4.260	1.326
Iris-virginica	125.5	6.588	2.974	5.552	2.026

In [22]: `data.Species.mode()`

Out[22]:

```
0    Iris-setosa
1    Iris-versicolor
2    Iris-virginica
Name: Species, dtype: object
```

In [23]: `data.SepalWidthCm.std()`

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In [20]: `data.groupby('Species').count()`

Out[20]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Species					
Iris-setosa	50	50	50	50	50
Iris-versicolor	50	50	50	50	50
Iris-virginica	50	50	50	50	50

In [21]: `data.groupby('Species').mean()`

Out[21]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Species					
Iris-setosa	25.5	5.006	3.418	1.464	0.244
Iris-versicolor	75.5	5.936	2.770	4.260	1.326
Iris-virginica	125.5	6.588	2.974	5.552	2.026

In [22]: `data.Species.mode()`

Out[22]:

```
0    Iris-setosa
1    Iris-versicolor
2    Iris-virginica
Name: Species, dtype: object
```

In [23]: `data.SepalWidthCm.std()`

Out[23]: 0.4335943113621737

In [24]: `data.SepalLengthCm.std()`

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In [20]: `data.groupby('Species').count()`

Out[20]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Iris-setosa	50	50	50	50	50
Iris-versicolor	50	50	50	50	50
Iris-virginica	50	50	50	50	50

In [21]: `data.groupby('Species').mean()`

Out[21]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Iris-setosa	25.5	5.006	3.418	1.464	0.244
Iris-versicolor	75.5	5.936	2.770	4.260	1.326
Iris-virginica	125.5	6.588	2.974	5.552	2.026

In [22]: `data.Species.mode()`

Out[22]:

```
0    Iris-setosa
1    Iris-versicolor
2    Iris-virginica
Name: Species, dtype: object
```

In [23]: `data.SepalWidthCm.std()`

Out[23]: 0.4335943113621737

In [24]: `data.SepalLengthCm.std()`

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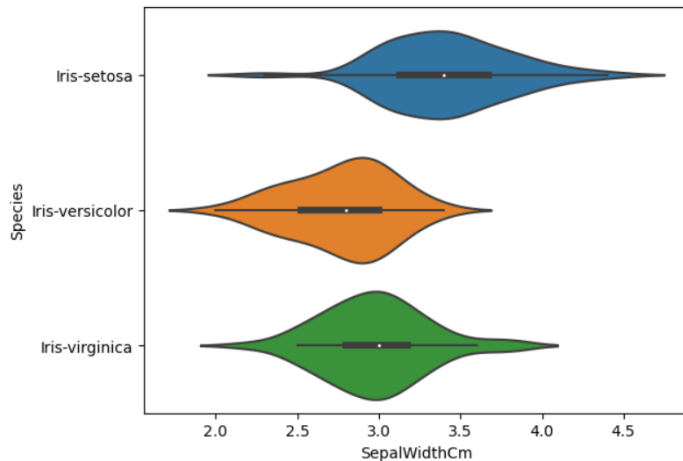
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In [24]: `data.SepalLengthCm.std()`

Out[24]: 0.8280661279778629

In [25]: `import seaborn as sns`
`sns.violinplot(x="SepalWidthCm", y="Species", data=data)`

Out[25]: <Axes: xlabel='SepalWidthCm', ylabel='Species'>



In [26]: `numeric_df = data.select_dtypes(include=[np.number])` # Select numeric columns

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